Computer Science & Information Technology

Discrete Mathematics

DPP: 3

Set Theory and Algebra

Q1 Let A= {a, b, c, d, e}.
Cardinality of the equivalence relation on set A induced by partition {{a, c}, {d}, {b, e}} of set A is_____

Q2 Let A is a finite set such that |A |= 5, Then cardinality of largest partial order relation possible on set A is_____

Q3 Let A = {a, b, c, d}
The number of total order relations possible on set A are_____

Q4 Let A = {1, 2, 3, 4, 5, 6}.
The number of partitions of set A such that equivalence relations induced by those partitions will contain exactly 12 ordered pairs are_____

Q5 Let R and S are two equivalence relation on set A. Which of the following is true?
(A) R ∪ S is also an equivalence relation on set A.
(B) R ∩ S is also an equivalence relation on set A.
(C) Both R ∪ S as well as R ∩ S are equivalence relation on set A.
(D) Neither R ∪ S nor R ∩ S are equivalence relation on set A.

Q6 Consider the following POSET
({{1}, {2}, {4}, {1, 2}, {1, 4}, {2, 4}, {3, 4}, {1, 3, 4}, {2, 3, 4}} , ⊆)
Let X is the number of minimal elements in the POSET and Y is the number of maximal elements in the POSET,
Then |X-Y| is_____

Q7 Consider the following POSET: ({2,3,4,9,18} , |)
Which of the following is/are true with respect to above POSET?
(A) Maximum element does not exist in the POSET.
(B) Exactly four pairs of elements in the above POSET are not comparable.
(C) Least upper bound of 3 and 9 is 18.
(D) Number of minimal elements and number of maximal elements are same.

Q8 Let A be a set with |A| = n, and let R be an equivalence relation on set A with |R| = r. Which of the following is TURE?
(A) r – n will always be even.
(B) r – n will always be odd.
(C) r – n may be zero.
(D) r – n may be 1.

Answer Key

Q1	9~9
Q2	15~15
Q3	24~24
Q4	35~35
Q5	(B)
Q6	0~0
Q7	(A, D)
Q8	(A, C)

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Hints & Solutions

Q1 Text Solution:
Cardinality of the equivalence relation on set A induced by partition {{a, c}, {d}, {b, e}} of set A is 9.

Q2 Text Solution:
The cardinality of largest partial order relation possible on set A is 15.

Q3 Text Solution:
The number of total order relations possible on set A are 24.

Q4 Text Solution:
The number of partitions of set A such that equivalence relations induced by those partitions will contain exactly 12 ordered pairs are 35.

Q6 Text Solution:
The value of |X-Y| is 0.

Q7 Text Solution:
Maximum element does not exist in the POSET.
Number of minimal elements and number of maximal elements are same.

Q8 Text Solution:
r – n will always be even.
r – n may be zero.

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